

Energy Efficiency in Railway Scheduling

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Outline

Introduction: Energy Efficiency Problem

Baseline ASP Model

Motivation for Energy Optimization

Energy Encoding Mechanics

Energy Costs of Actions

Search Optimization and Heuristics

Results and Comparison with Baseline Model

Conclusion

Energy Efficiency Problem

Baseline ASP Model

- ▶ Establishes rules to handle essential train behavior
 - ▶ Directions
 - ▶ Movements
 - ▶ Transitions
 - ▶ Collision and Reachability Constraints
 - ▶ Schedule Optimization

Directions and Movements

% Directions

dir(s).

dir(n).

dir(w).

dir(e).

% Movement Operation

moving(move_forward).

moving(move_left).

moving(move_right).

% Movement on Grid

delta(A,s,1,0) :- moving(A).

delta(A,n,-1,0) :- moving(A).

delta(A,e,0,1) :- moving(A).

delta(A,w,0,-1) :- moving(A).

Transitions

```
% Transition facts: To represent different track types
% ID : track ID
% Dir1: pointing direction of agent
% Act : associated action with movement towards exit
% Dir2: exit direction
transition(ID, Dir1, Act, Dir2).
```

Collision Constraints

```
% Trains can not occupy the same cell at the same time  
:- stay(ID1,X,Y,T),  
   stay(ID2,X,Y,T),  
   ID1 < ID2.
```

```
% Trains can not swap their position  
:- stay(ID1,X1,Y1,T),  
   stay(ID2,X2,Y2,T),  
   stay(ID1,X2,Y2,T+1),  
   stay(ID2,X1,Y1,T+1),  
   ID1 < ID2.
```

Reachability Constraints

```
% define states (cell, time, direction),  
% which are reachable by a train  
reachable(ID,T,A,X2,Y2,T2,Dir2) :-  
    A != wait,  
    position(ID,(X,Y),T,Dir1),  
    cell((X,Y),Cell),  
    transition(Cell,Dir1,A,Dir2),  
    delta(A,Dir2,DX,DY),  
    X2 = X + DX,  
    Y2 = Y + DY,  
    speed(ID,S),  
    T2 = T + S,  
    time(T2).  
  
reachable(ID,T,wait,X,Y,T+1,Dir) :-  
    position(ID,(X,Y),T,Dir),  
    time(T+1).
```


Reachability Constraints

```
% Choose exactly one valid transition at each time step
1 { position(ID,(X2,Y2),T2,Dir2)
    : reachable(ID,T,A,X2,Y2,T2,Dir2)
} 1 :- position(ID,(_,_),T,_),
    not reached(ID,T).

% Track the time interval a train spends inside a cell.
% This is to make enforcing collision constraints easier.
chosen(ID,T,A,X,Y,X2,Y2,T2) :-
    position(ID,(X,Y),T,_),
    position(ID,(X2,Y2),T2,_),
    reachable(ID,T,A,X2,Y2,T2,_).

% a train reached its target if their position matches
reached(ID,T) :- position(ID,(X,Y),T,_),
    end(ID,(X,Y),_).
```

Schedule Optimization

```
% stop searching if train is not
% in time to reach its target anymore
:- position(ID,(X,Y),T,_),
   end(ID,(EX,EY),Arr),
   speed(ID,InvSpd),
   T + (|X-EX| + |Y-EY|)*InvSpd > Arr.

% the arrival times should be minimized
reached_earlier(ID,T) :- reached(ID,T),
                        reached(ID,T0), T0 < T.
arrival(ID,T) :- reached(ID,T),
                 not reached_earlier(ID,T).

% minimize summed arrival times
#minimize { T@2,ID : arrival(ID,T) }.
```

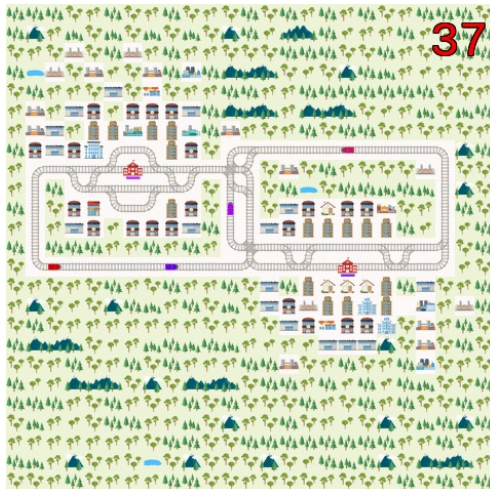


Figure: Baseline Encoding in Action (at timestep 37)

Motivation

Energy Encoding Mechanics

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